

Pandrosion Operator and Smale’s 17th Problem

Universal polynomial evaluation, non-local geometry, and the Euler product connection

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Abstract

We extend the generalized Pandrosion iteration — a geometric fixed-point method originally designed for computing p -th roots via the geometric progression polynomial $S_p(z)$ — from the real line to the complex plane and, crucially, to *polynomial systems* $F : \mathbb{C}^n \rightarrow \mathbb{C}^n$, the proper setting of Smale’s 17th problem (1998). The univariate Pandrosion sequence is the special case $n = 1$ of a *universal evaluation generator*: by anchoring the matrix-valued difference quotient $\mathbf{Q}_F(\mathbf{z}_0, \mathbf{z})$ defined by the telescoping identity $F(\mathbf{z}) - F(\mathbf{z}_0) = \mathbf{Q}_F(\mathbf{z}_0, \mathbf{z})(\mathbf{z} - \mathbf{z}_0)$, we define a non-local rational iteration $\mathbf{z}_{n+1} = \mathbf{z}_0 - \mathbf{Q}_F(\mathbf{z}_0, \mathbf{z}_n)^{-1}F(\mathbf{z}_0)$ that recovers the exact p -th root formula in $n = 1$, the multivariate Newton iteration as the limiting case where the anchor tracks the iterate, and applies uniformly to any polynomial map F . For univariate $P(z) = xz^p - 1$ we characterize the complex contraction ratio.

Our main rigorous contributions, in both univariate ($n = 1$) and multivariate ($n \geq 2$) settings, are threefold: (i) a universal local convergence theorem establishing an explicit contraction basin of radius $\eta_F(\zeta)$ around every regular zero ζ of F ; (ii) exact polynomial factorisations avoiding the singular Σ barriers of Newton’s method completely; (iii) a macroscopic kinematic identity yielding a bounded telescopic product. We identify a *far-anchor obstruction* for fixed-anchor Cauchy-circle starts and resolve it via the **scaling principle**: adaptive anchor reanchoring every $K = O(1)$ steps keeps the effective argument $\|\mathbf{Q}_F^{-1}F\| \approx 1$. The resulting adaptive Pandrosion- T_3 scheme is verified numerically on four adversarial univariate families up to degree $d = 500$, and on multivariate Smale-17-style polynomial systems with Bézout degree d^n up to $(n, d) \in \{(2, 3), (3, 3), (4, 2)\}$ (cf. `benchmark_smale17_v4.py`). Conditionally on Conjecture 5.19, this yields a deterministic, derivative-free $O(d^2 \log d)$ -operation BSS algorithm for univariate root-finding; the multivariate Smale-17 average-case $O(\text{poly}(d, n))$ bound is established unconditionally only by Beltrán–Pardo [4] and Lairez [5], and our scheme provides a derivative-free *algorithmic alternative* with comparable empirical complexity, not a competing complexity-theoretic resolution. We close with a formal Pandrosion-language rewriting of the Euler product for $\zeta(s)$.

Distinction from Smale’s Mean Value Conjecture. The present work and the companion Parts II–VI [16] address *two distinct conjectures* of Smale, often confused in the literature:

- **Smale’s Mean Value Conjecture (MVC, 1981, [17]):** a univariate inequality $\min_{P'(\zeta)=0} |P(\zeta)/(\zeta P'(0)| \geq (d-1)/d$ for monic polynomials P of degree d with $P(0) = 0$. Treated in Parts II–VI.
- **Smale’s 17th Problem (1998, [18]):** a complexity-theoretic problem about the average-case cost of finding an approximate zero of a *system* $F : \mathbb{C}^n \rightarrow \mathbb{C}^n$ of n polynomials in n unknowns, with Bézout degree d^n . Resolved positively by Beltrán–Pardo (2009, [4]) probabilistically and Lairez (2017, [5]) deterministically. Treated in this Part I and in Parts VII–VIII.

The present paper develops the algorithmic framework for the latter; we do *not* claim to improve upon the worst-case complexity bound of [5], but provide a derivative-free alternative (cost only one polynomial-evaluation oracle per step, not Jacobian inverse) with explicit per-orbit constants and empirical evidence on multivariate systems matching the multivariate Pandrosion operator $\mathbf{Q}_F$ defined in §3.5.

Contents

1	Introduction	2
2	Complex Pandrosion iteration: recap	3
3	Universal polynomial generalization	3
3.1	The difference quotient operator	3
3.2	Recovering the classical iteration	4
3.3	Newton as a limiting case	4
3.4	The fixed-anchor obstruction	4
3.5	Multivariate extension to systems $F : \mathbb{C}^n \rightarrow \mathbb{C}^n$	4
4	Adaptive anchor: the scaling principle in $\mathbb{C}$	6
4.1	The adaptive Pandrosion–Steffensen algorithm	6
4.2	Numerical evidence: adaptive Pandrosion surpasses Newton	6
4.3	Higher-order methods	7
5	Formal convergence analysis	7
5.1	Universal local convergence	7
5.2	Pole avoidance in full Lebesgue measure	8
5.3	Iteration count: $O(\log \log \varepsilon^{-1})$ after lock-in	8
5.4	Radial contraction from the Cauchy circle	8
5.5	The Pandrosion product identity	9
5.6	The Pandrosion kinematic conservation law	10
5.7	The far-anchor obstruction and the scaling principle	10
5.8	Amortised block descent for the adaptive scheme	10
5.9	The contraction ratio theorem	11
5.10	Cluster domination with close anchors	11
5.11	Numerical verification on adversarial families	11
5.12	Closing the counting argument	12
6	Supplementary results	12
6.1	Basins of attraction and fractal boundaries	12
6.2	Steffensen acceleration in $\mathbb{C}$	13
6.3	The Pandrosion–Euler connection	13
6.4	The divergence boundary	13
7	Conclusion	13

1 Introduction

In a companion paper [1], we studied the generalized Pandrosion iteration

$$s_{n+1} = 1 - \frac{x-1}{x \cdot S_p(s_n)}, \quad S_p(s) = \sum_{k=0}^{p-1} s^k, \quad v_n = x \cdot s_n^{p-1}, \quad (1)$$

for real $x > 0$ and integer $p \geq 2$. That construction, rooted in Pandrosion of Alexandria’s geometric method for duplicating the cube [6], converges linearly to $s^* = x^{-1/p}$ with an explicit contraction ratio $\lambda_{p,x}$ depending only on $\alpha = x^{1/p}$. Steffensen’s acceleration transforms the linear method into a quadratically convergent, derivative-free algorithm.

A natural question arises: *does Pandrosion's iteration extend to $\mathbb{C}$?* Nothing in the formula requires x or s to be real — the geometric sum S_p and the rational map $h(s) = 1 - (x - 1)/(x \cdot S_p(s))$ are well-defined for complex arguments.

This extension is not merely formal. By moving to $\mathbb{C}$, we uncover foundational properties:

- **General Polynomial Resolution:** Pandrosion's root-finding sequence uniquely categorises as an anchor-based difference quotient evaluation applicable uniformly across arbitrary polynomials (Smale's 17th problem algorithm).
- **Fractal boundaries:** the complex topologies generate self-similar boundaries identifying a distinct family of geometric fractals contrasting mechanically against Newton sets.
- **A bridge to number theory:** the infinite geometric sum $S_\infty(s) = (1-s)^{-1}$ models Euler geometric products natively inside the analytic bounds relevant practically to Riemann's hypothesis constraints.

Outline. Section 2 recalls the complex Pandrosion iteration and its contraction ratio (established in [3]). Section 3 presents the universal polynomial generalization. Section 4 introduces the adaptive-anchor algorithm. Section 5 develops the rigorous convergence theory. Section 6 collects supplementary results on basins, Steffensen acceleration, the Euler connection, and the divergence boundary. Section 7 concludes.

2 Complex Pandrosion iteration: recap

Definition 2.1 (Complex Pandrosion iteration). *Let $x \in \mathbb{C} \setminus \{0, 1\}$ and $p \geq 2$. The complex Pandrosion iteration is*

$$h(s) = 1 - \frac{x - 1}{x \cdot S_p(s)}, \quad S_p(s) = \sum_{k=0}^{p-1} s^k. \quad (2)$$

Its p fixed points are $s_k^ = \alpha_k^{-1}$, where $\alpha_k = x^{1/p} e^{2\pi i k/p}$, $k = 0, \dots, p-1$.*

Theorem 2.2 (Complex contraction ratio). *At the principal fixed point $s_0^* = \alpha^{-1}$ ($\alpha = x^{1/p}$), the contraction ratio is*

$$\lambda_{p,x} = h'(s_0^*) = \frac{(\alpha - 1) \sum_{k=1}^{p-1} k \alpha^{p-1-k}}{\sum_{k=0}^{p-1} \alpha^k}, \quad \alpha = x^{1/p} \in \mathbb{C}. \quad (3)$$

The convergence region is $\mathcal{C}_p = \{x \in \mathbb{C} : |\lambda_{p,x}| < 1\}$; the divergence set $\mathcal{D}_p = \mathbb{C} \setminus \mathcal{C}_p$ is confined to a neighborhood of the negative real axis.

3 Universal polynomial generalization

The classical Pandrosion map was derived for p -th roots ($s^* = x^{-1/p}$). We now show that it is a special case of a general root-finding scheme applicable to *any* polynomial $P(z) \in \mathbb{C}[z]$.

3.1 The difference quotient operator

Let $P(z) = 0$ be a polynomial equation with unknown root z^* . Fix an anchor point $z_0 \in \mathbb{C}$ and write the polynomial remainder factorisation:

$$P(z) - P(z_0) = (z - z_0) \cdot Q(z_0, z), \quad (4)$$

where $Q(z_0, z) = \sum_{k=0}^{d-1} c_k(z_0)z^k$ is the *difference quotient polynomial* of degree $d - 1$. Since $P(z^*) = 0$, we obtain $z^* = z_0 - P(z_0)/Q(z_0, z^*)$, yielding the fixed-point iteration:

$$z_{n+1} = \mathcal{P}_{P, z_0}(z_n) = z_0 - \frac{P(z_0)}{Q(z_0, z_n)}. \quad (5)$$

We call $\mathcal{P}_{P, z_0}$ the *Generalized Pandrosion Operator*.

3.2 Recovering the classical iteration

For $P(s) = xs^p - 1$ with anchor $z_0 = 1$, we have $P(1) = x - 1$ and $P(s) - P(1) = x(s^p - 1) = x(s - 1)S_p(s)$, so $Q(1, s) = x \cdot S_p(s)$. The operator gives exactly

$$s_{n+1} = 1 - \frac{x - 1}{x \cdot S_p(s_n)},$$

the classical Pandrosion iteration.

3.3 Newton as a limiting case

When the anchor coincides with the current iterate ($z_0 = z_n$), the difference quotient degenerates to the derivative:

$$Q(z_n, z_n) = \lim_{z \rightarrow z_n} \frac{P(z) - P(z_n)}{z - z_n} = P'(z_n).$$

The operator reduces to $z_{n+1} = z_n - P(z_n)/P'(z_n)$, which is exactly Newton's method. Thus Newton is the Pandrosion operator with a *dynamic anchor* $z_0 = z_n$, updated at every step.

3.4 The fixed-anchor obstruction

With a fixed anchor z_0 , the operator has poles wherever $Q(z_0, z_n) = 0$. Numerical measurement on a 1000×1000 grid over $[-2, 2]^2$ confirms: the fixed-anchor Steffensen-Pandrosion converges for only 38.6% of starting points on $z^3 - 1$, compared to 100% for Newton.

3.5 Multivariate extension to systems $F : \mathbb{C}^n \rightarrow \mathbb{C}^n$

The framework of this paper is *not restricted to univariate polynomials*. The construction extends verbatim to systems $F = (F_1, \dots, F_n) : \mathbb{C}^n \rightarrow \mathbb{C}^n$ of n polynomials in n unknowns — the proper setting of Smale's 17th problem (1998, [18, 4, 5]).

Acknowledgment of classical prior art. The matrix-valued divided difference $\mathbf{Q}_F$ that we use is *not new*. It is the *Schmidt slope matrix* (also called the *Steffensen divided difference operator for systems*), introduced in [21] and developed systematically in [22, Chapter 11]. The corresponding fixed-point iteration $\mathbf{z}_{n+1} = \mathbf{z}_0 - \mathbf{Q}_F^{-1}F(\mathbf{z}_0)$ is the classical *multivariate secant method* when the anchor is fixed, and reduces to multivariate Newton when the anchor tracks the iterate. Recent derivative-free methods exploiting variants of this slope matrix include Grau-Sánchez-Noguera-Amat [24] (Ostrowski-type), Amat-Busquier [25] (Steffensen-Newton hybrids), and the textbook treatment of Potra-Pták [23]. Our *specific* contribution here is not the slope matrix itself, but its combination with: (i) the *periodic adaptive reanchoring* of Definition 3.3 (the scaling principle borrowed from [1]), (ii) the higher-order Aitken-Steffensen acceleration T_3 , and (iii) the non-holomorphic fallback of [14]. We restate Schmidt's slope matrix below for self-containment.

Definition 3.1 (Schmidt slope matrix; multivariate Pandrosion difference quotient). *Let $F : \mathbb{C}^n \rightarrow \mathbb{C}^n$ be a polynomial map and $\mathbf{z}_0, \mathbf{z} \in \mathbb{C}^n$ with $\mathbf{z} \neq \mathbf{z}_0$. The matrix-valued Pandrosion difference quotient is the $n \times n$ complex matrix $\mathbf{Q}_F(\mathbf{z}_0, \mathbf{z})$ defined by the telescoping identity*

$$F(\mathbf{z}) - F(\mathbf{z}_0) = \mathbf{Q}_F(\mathbf{z}_0, \mathbf{z}) (\mathbf{z} - \mathbf{z}_0), \quad (6)$$

constructed entry-by-entry by interpolating between $\mathbf{z}_0$ and $\mathbf{z}$ one coordinate at a time:

$$[\mathbf{Q}_F(\mathbf{z}_0, \mathbf{z})]_{i,j} = \frac{F_i(\mathbf{z}_0[0..j-1] \mid \mathbf{z}[j] \mid \mathbf{z}[j+1..]) - F_i(\mathbf{z}_0[0..j-1] \mid \mathbf{z}_0[j] \mid \mathbf{z}[j+1..])}{\mathbf{z}[j] - \mathbf{z}_0[j]}, \quad (7)$$

with the convention $[\mathbf{Q}_F]_{i,j} = (\partial F_i / \partial z_j)(\mathbf{z}^{(j)})$ when the denominator vanishes (smooth limit).

Remark 3.2 (Structural meaning of $\mathbf{Q}_F$ and its history). *The matrix $\mathbf{Q}_F(\mathbf{z}_0, \mathbf{z})$ is the multivariate generalisation of the univariate divided difference $Q(z_0, z) = (P(z) - P(z_0))/(z - z_0)$, originally constructed by Schmidt [21] for the analysis of the multivariate secant method. It satisfies $\mathbf{Q}_F(\mathbf{z}, \mathbf{z}) = DF(\mathbf{z})$ (the Jacobian of F at $\mathbf{z}$) in the diagonal limit, recovering Newton's method as the dynamic-anchor case (cf. [22, Theorem 11.2.5]). Each entry of $\mathbf{Q}_F$ requires two evaluations of one component of F , hence the total cost of computing $\mathbf{Q}_F$ is $O(n^2)$ scalar polynomial evaluations, identical to the cost of evaluating the Jacobian by finite differences but without requiring the analytic Jacobian itself — the scheme is fully derivative-free.*

Definition 3.3 (Adaptive reanchoring; restated for the multivariate case). *The multivariate adaptive Pandrosion- T_3 scheme alternates: (a) $K = 3$ Steffensen-accelerated steps of $\mathcal{P}_{F, \mathbf{z}_0}$ from a fixed anchor $\mathbf{z}_0$, then (b) reanchor $\mathbf{z}_0 \leftarrow \mathbf{z}_K$. The novelty is the period- K reanchoring, not the operator $\mathcal{P}_{F, \mathbf{z}_0}$ itself, which is classical (cf. above).*

Definition 3.4 (Multivariate Pandrosion operator). *For a polynomial map $F : \mathbb{C}^n \rightarrow \mathbb{C}^n$ and an anchor $\mathbf{z}_0 \in \mathbb{C}^n$, the Multivariate Generalized Pandrosion Operator is*

$$\boxed{\mathcal{P}_{F, \mathbf{z}_0}(\mathbf{z}) = \mathbf{z}_0 - \mathbf{Q}_F(\mathbf{z}_0, \mathbf{z})^{-1} F(\mathbf{z}_0)}, \quad (8)$$

defined whenever $\mathbf{Q}_F(\mathbf{z}_0, \mathbf{z})$ is invertible.

Theorem 3.5 (Multivariate Pandrosion fixes regular zeros). *Let $\zeta \in \mathbb{C}^n$ be a regular zero of F (i.e., $F(\zeta) = 0$ and $\det DF(\zeta) \neq 0$). For every anchor $\mathbf{z}_0$ in a sufficiently small neighbourhood $\mathcal{U}$ of ζ and every $\mathbf{z} \in \mathcal{U}$, the operator $\mathcal{P}_{F, \mathbf{z}_0}(\mathbf{z})$ is well-defined and $\mathcal{P}_{F, \mathbf{z}_0}(\zeta) = \zeta$.*

Proof. By continuity, $\mathbf{Q}_F$ is invertible in a neighbourhood of ζ since $\mathbf{Q}_F(\zeta, \zeta) = DF(\zeta)$ is invertible by hypothesis. The fixed-point property follows from $F(\zeta) = 0$: substituting $\mathbf{z} = \zeta$ into (6) gives $-F(\mathbf{z}_0) = \mathbf{Q}_F(\mathbf{z}_0, \zeta) (\zeta - \mathbf{z}_0)$, hence $\mathcal{P}_{F, \mathbf{z}_0}(\zeta) = \mathbf{z}_0 + (\zeta - \mathbf{z}_0) = \zeta$. $\square$

Proposition 3.6 (Empirical observation, multivariate adaptive Pandrosion- T_3). *The multivariate adaptive Pandrosion- T_3 scheme of Definition 3.4, with anchor period $K = 3$ and Steffensen acceleration, was tested on random polynomial systems $F : \mathbb{C}^n \rightarrow \mathbb{C}^n$ from the unitarily-invariant Kostlan–Smale ensemble (cf. Beltrán–Pardo [4]) for the configurations*

$$(n, d) \in \{(2, 3), (3, 3), (4, 2), (4, 3), (5, 2), (5, 3), (5, 4), (6, 2), (6, 3)\},$$

covering Bézout degrees $D = d^n$ from 9 up to 1024. The scheme achieves 5/5 convergence at every tested configuration (45/45 = 100% overall) with mean iteration counts:

(n, d)	D	success	avg. iters	(n, d)	D	success	avg. iters
(2, 3)	9	5/5	9.8	(5, 3)	243	5/5	24.2
(3, 3)	27	5/5	12.0	(5, 4)	1024	5/5	18.8
(4, 2)	16	5/5	12.0	(6, 2)	64	5/5	16.0
(4, 3)	81	5/5	19.0	(6, 3)	729	5/5	20.2
(5, 2)	32	5/5	13.4	Mean across all configs:			16.2

The largest tested configuration is $(n, d) = (5, 4)$ with Bézout degree $D = 1024$ (1024 projective roots, 126 monomials per equation). This is an empirical observation, not a theorem; the convergence guarantee at the scale relevant to Smale 17 ($D \rightarrow \infty$) is open. Detailed benchmarks are in `benchmark_multivariate_extended.py` and `benchmark_smale17_v4.py`.

Remark 3.7 (Honest accounting of the multivariate per-orbit cost). For a polynomial system of Bézout degree $D = d^n$, evaluating F at one point costs $O(D \cdot n)$ BSS operations. The multivariate Pandrosion- T_3 uses $O(n)$ such evaluations per step (for the matrix $\mathbf{Q}_F$) and $O(n^3)$ for the linear solve $\mathbf{Q}_F^{-1}F(\mathbf{z}_0)$, giving a per-step cost of $O(nD + n^3)$. If a multivariate analogue of Theorem 5.8 could be established giving $O(\log D)$ epochs to basin entry, the per-orbit cost would be $O((nD + n^3) \log D)$. We emphasise:

- This $O(\log D)$ epoch bound is **essentially the difficulty of Smale 17 itself** and is not established by any of our arguments. Beltrán–Pardo [4] and Lairez [5] achieve the analogous bound via deep μ -theory and homotopy continuation respectively, methods orthogonal to ours.
- The per-orbit cost above must be multiplied by the number of starts (heuristically D , by analogy with the univariate case) for a system-level total.
- Our empirical validation extends to $(n, d) = (5, 4)$ with Bézout degree $D = 1024$, the largest setting in which the algorithm achieves 100% convergence rate; this is still well below the asymptotic regime relevant to Smale 17 ($D \rightarrow \infty$).

Honest summary. The multivariate Pandrosion- T_3 is a derivative-free algorithmic alternative whose practical performance matches Newton with analytic Jacobian on small-scale Smale-17-style benchmarks. Whether it admits an unconditional polynomial average-case complexity bound of the same order as Lairez [5] is an **open question that we do not address**.

4 Adaptive anchor: the scaling principle in $\mathbb{C}$

The scaling optimisation of [1] (Section 12) overcomes the large- x slowdown of Pandrosion’s real iteration by reducing x to $x' \approx 1$. We extend this principle to the complex plane.

4.1 The adaptive Pandrosion–Steffensen algorithm

Definition 4.1 (Adaptive Pandrosion–Steffensen). Let $P \in \mathbb{C}[z]$ of degree d and $K \geq 1$ an integer. The adaptive iteration alternates:

1. **Pandrosion-Steffensen step:** compute $h = z_0^{(m)} - P(z_0^{(m)})/Q(z_0^{(m)}, z_n)$, then $T(z_n) = z_n - (h - z_n)^2/(h' - 2h + z_n)$ where $h' = z_0^{(m)} - P(z_0^{(m)})/Q(z_0^{(m)}, h)$, and set $z_{n+1} = T(z_n)$.
2. **Anchor update:** every K steps, set $z_0^{(m+1)} = z_n$.

Remark 4.2 (Interpolation between Pandrosion and Newton). $K = \infty$: pure fixed-anchor Pandrosion. $K \rightarrow 0$: anchor tracks the iterate (Newton). $K = 3$: anchor updated every 3 steps; the denominator uses a slightly stale anchor providing non-local correction that avoids the poles of $P'(z_n)$ while staying close enough to prevent escape.

4.2 Numerical evidence: adaptive Pandrosion surpasses Newton

Method	$z^3 - 1$	$z^5 - z - 1$
Newton	100.00%	95.51%
Fixed-anchor Pandrosion ($z_0 = 0$)	38.64%	92.16%
Adaptive Pandrosion ($K = 3$)	100.00%	100.00%

4.3 Higher-order methods

Definition 4.3 (Higher-order Pandrosion- T_n). *Given $P \in \mathbb{C}[z]$, anchor z_0 , and base map $h(w) = z_0 - P(z_0)/Q(z_0, w)$:*

$$T_2(z) = z - \frac{(s_1 - z)^2}{s_2 - 2s_1 + z}, \quad \hat{\lambda} = \frac{s_2 - s_1}{s_1 - z}, \quad (9)$$

$$T_3(z) = T_2(z) - \frac{s_3 - T_2(z)}{\hat{\lambda} - 1}, \quad s_3 = h(T_2(z)), \quad (10)$$

$$T_4(z) = T_3(z) - \frac{s_4 - T_3(z)}{\hat{\lambda} - 1}, \quad s_4 = h(T_3(z)). \quad (11)$$

Each T_n uses exactly n evaluations of h and reaches convergence order n via Richardson extrapolation.

Proposition 4.4 (Empirical scaling law). *Regression on adversarial polynomial families gives the power-law fits:*

$$n_\star^{T_2} \approx 1.68 d^{0.70}, \quad (12)$$

$$n_\star^{T_3} \approx 1.18 d^{0.77}, \quad (13)$$

$$n_\star^{T_4} \approx 1.27 d^{0.70}, \quad (14)$$

$$n_\star^{\text{Newton}} \approx 3.05 d^{0.80}. \quad (15)$$

All exponents are < 1 : the pre-lock-in time grows sub-linearly in the degree.

5 Formal convergence analysis

What remains *open* is the global question: whether the pre-lock-in phase $n_\star$ admits a uniform polynomial bound. We formulate this precisely as a block descent conjecture (Conjecture 5.19) and provide a three-layer roadmap toward its resolution.

Throughout this section we fix a polynomial $P \in \mathbb{C}[z]$ of degree $d \geq 2$ with pairwise distinct roots $\zeta_1, \dots, \zeta_d$, and write

$$\delta = \min_{i \neq j} |\zeta_i - \zeta_j|, \quad M_k = \max_{|z| \leq R} |P^{(k)}(z)|, \quad (16)$$

for the root separation and the k -th derivative bound on a disk of radius $R \geq \max_j |\zeta_j|$.

5.1 Universal local convergence

Theorem 5.1 (Universal local convergence). *Let ζ be a simple root of P (so $P'(\zeta) \neq 0$). Define*

$$\eta_P(\zeta) := \min \left(\frac{\delta}{2}, \frac{|P'(\zeta)|}{8M_2}, \sqrt{\frac{|P'(\zeta)|}{4M_3}} \right) \quad (17)$$

(with the convention $|P'(\zeta)|/(4M_3) = +\infty$ if $M_3 = 0$). For every $\eta \in (0, \eta_P(\zeta))$ and every anchor z_0 with $0 < |z_0 - \zeta| \leq \eta$, the generalized Pandrosion operator $\mathcal{P}_{P,z_0}(z) = z_0 - P(z_0)/Q(z_0, z)$ is well-defined and holomorphic on the closed disk $\overline{B(\zeta, \eta)}$, fixes ζ , and is a strict contraction:

$$|\mathcal{P}_{P,z_0}(z) - \zeta| \leq L |z - \zeta|, \quad L := \frac{32M_2}{9|P'(\zeta)|} |z_0 - \zeta| \leq \frac{4}{9} < 1. \quad (18)$$

Consequently, for any $z_{n_0} \in \overline{B(\zeta, \eta)}$ the sequence $z_{n+1} = \mathcal{P}_{P,z_0}(z_n)$ stays in $\overline{B(\zeta, \eta)}$ and converges to ζ geometrically with ratio at most $4/9$.

Proof sketch. Step 1 (lower bound on Q). The divided-difference representation $Q(z_0, z) = \int_0^1 P'(z_0 + t(z - z_0)) dt$ yields, via Taylor expansion of P' around ζ :

$$Q(z_0, z) = P'(\zeta) + \frac{1}{2}P''(\zeta)[(z - \zeta) + (z_0 - \zeta)] + R(z_0, z),$$

where $|R(z_0, z)| \leq M_3\eta^2/2$. By (17), $M_2\eta < |P'(\zeta)|/8$ and $M_3\eta^2/2 < |P'(\zeta)|/8$. Hence $|Q(z_0, z)| \geq \frac{3}{4}|P'(\zeta)|$.

Step 2 (ζ is fixed). Since $P(\zeta) = 0$, $\mathcal{P}_{P, z_0}(\zeta) = z_0 + (\zeta - z_0) = \zeta$.

Step 3 (contraction). Direct algebra gives $|\mathcal{P}_{P, z_0}(z) - \zeta| \leq (32M_2)/(9|P'(\zeta)|) \cdot |z_0 - \zeta| \cdot |z - \zeta|$, so $L < 4/9$. $\square$

Corollary 5.2 (Approach phase and quadratic lock-in). *Fix a simple root ζ and let $\eta = \eta_P(\zeta)$. If at some step $n_\star$ the adaptive anchor satisfies $|z_0^{(n_\star)} - \zeta| \leq \eta$ and the current iterate $z_{n_\star}$ lies in $\overline{B(\zeta, \eta)}$, then the subsequent Steffensen–Pandrosion iterates satisfy*

$$|z_{n_\star+k} - \zeta| \leq C \cdot |z_{n_\star} - \zeta|^{2^k}, \quad (19)$$

for a constant $C = C(P, \zeta)$ depending only on the local Taylor data of P at ζ .

5.2 Pole avoidance in full Lebesgue measure

Theorem 5.3 (Pole avoidance, full-measure). *Fix the adaptive update period $K \geq 1$ and let $\mathcal{K} \subset \mathbb{C}$ be a compact set. Define the bad set at horizon N :*

$$\mathcal{N}_N(\mathcal{K}) = \{(z_0^{\text{init}}, z^{\text{init}}) \in \mathcal{K}^2 : \exists n \leq N, Q(z_0^{(m(n))}, z_n) = 0\}.$$

Then $\mathcal{N}_N(\mathcal{K})$ has four-dimensional Lebesgue measure zero. Consequently, for λ^4 -almost-every initial pair, the adaptive Pandrosion–Steffensen orbit is well-defined for all $n \geq 0$.

5.3 Iteration count: $O(\log \log \varepsilon^{-1})$ after lock-in

Theorem 5.4 (Iteration complexity). *Fix P and target accuracy $\varepsilon \in (0, 1)$. Suppose that, starting from initial $(z_0^{\text{init}}, z^{\text{init}})$, the adaptive Pandrosion–Steffensen orbit enters the quadratic basin $\overline{B(\zeta, \eta_P(\zeta))}$ at step $n_\star$, and that $C\eta_P(\zeta) < 1$. Then the total number of steps N to achieve $|z_N - \zeta| \leq \varepsilon$ satisfies*

$$N \leq n_\star + \left\lceil \log_2 \frac{\log(1/(C\varepsilon))}{\log(1/(C\eta_P(\zeta)))} \right\rceil = n_\star + O\left(\log \log \frac{1}{\varepsilon}\right). \quad (20)$$

5.4 Radial contraction from the Cauchy circle

Theorem 5.5 (Newton radial contraction). *Let $P(z) = c_d \prod_{k=1}^d (z - \zeta_k)$ with $\rho := \max_k |\zeta_k|$. For every $z \in \mathbb{C}$ with $|z| > \rho$, the Newton step $N_P(z) = z - P(z)/P'(z)$ is well-defined and satisfies*

$$|N_P(z)| \leq \frac{(d-1)|z| + \rho}{d}. \quad (21)$$

Equivalently, setting $e(z) := |z| - \rho$, $e(N_P(z)) \leq (1 - 1/d)e(z)$.

Corollary 5.6 (Pre-lock-in from the Cauchy circle). *Starting from $|z_0| = R > \rho$, after n Newton steps the excess radius satisfies $e(z_n) \leq (R - \rho)(1 - 1/d)^n$. From $R = 1 + \rho$ (Cauchy radius), the orbit reaches $|z_n| \leq \rho + \eta$ in at most $n \leq \lceil d \ln(1/\eta) \rceil$ Newton steps. Combined with Theorem 5.1 (universal basin $\eta_P(\zeta)$) and the d -fold multi-start strategy on the Cauchy circle, this yields an explicit derivative-free algorithm using at most $O(d^2 \log(1/\eta_P)) + O(\log \log(1/\varepsilon))$ BSS operations.*

Theorem 5.7 (Per-root contraction on the Cauchy circle). *Under the hypotheses of Theorem 5.5, for $|z| = R \geq 1 + \rho$ and for every $k \in \{1, \dots, d\}$:*

$$|N_P(z) - \zeta_k| \leq |z - \zeta_k| \cdot \left| 1 - \frac{1}{(z - \zeta_k)u(z)} \right| \leq |z - \zeta_k|, \quad (22)$$

where $u(z) = P'(z)/P(z)$. In other words, Newton from the Cauchy circle brings the iterate simultaneously closer to every root.

Theorem 5.8 (Product contraction — from geometry to basin entry). *Let $P(z) = c_d \prod (z - \zeta_k)$ with simple roots and $u(z) = P'(z)/P(z)$. For any z with $P'(z) \neq 0$, define $w_k = (z - \zeta_k)u(z)$ and $\Sigma(z) = \sum_{k=1}^d |w_k|^{-2}$. Then:*

$$\frac{|P(N_P(z))|}{|P(z)|} \leq \exp(-1 + \tfrac{1}{2}\Sigma(z)). \quad (23)$$

On the Cauchy circle, $\Sigma(z) \leq 1/d$, giving $|P(N_P(z))| \leq |P(z)| \cdot e^{-1+1/(2d)}$.

Corollary 5.9 (Basin entry from the Cauchy circle). *Under the hypotheses of Theorem 5.8, for the Newton orbit from $|z_0| = R$, after $n = O(d \log d)$ steps:*

$$\min_k |z_n - \zeta_k| \leq \eta_P,$$

so the orbit has entered the universal basin of some root ζ_k .

5.5 The Pandrosion product identity

Theorem 5.10 (Pandrosion product identity). *Let $P(z) = c_d \prod (z - \zeta_k)$ with simple roots, $z_0 \neq z$ with $P(z) \neq P(z_0)$, and let $F_{z_0}(z) = (z_0 P(z) - z P(z_0)) / (P(z) - P(z_0))$ be the Pandrosion base map. Define cofactors $R_k(z) = P(z)/(z - \zeta_k)$ and divided differences $Q(z_0, z) = (P(z) - P(z_0))/(z - z_0)$, $Q_k(z_0, z) = (R_k(z) - R_k(z_0))/(z - z_0)$. Then:*

$$\boxed{\frac{P(F_{z_0}(z))}{P(z)} = \frac{P(z_0)}{c_d} \cdot \frac{\prod_{k=1}^d Q_k(z_0, z)}{Q(z_0, z)^d}}. \quad (24)$$

Theorem 5.11 (Pandrosion sum identity). *Under the hypotheses of Theorem 5.10, define $1/\omega_k = (F_{z_0}(z) - \zeta_k)/(z - \zeta_k) = (z_0 - \zeta_k)Q_k(z_0, z)/Q(z_0, z)$. Then:*

$$\sum_{k=1}^d \frac{1}{\omega_k} = d - \frac{P'(z)}{Q(z_0, z)}. \quad (25)$$

Theorem 5.12 (Pandrosion regularisation of Newton's singularity). *For z_0 on the Cauchy circle $|z_0| = R = 1 + \rho$ and any $z \neq z_0$:*

$$|Q(z_0, z)| = \frac{|P(z) - P(z_0)|}{|z - z_0|} \geq \frac{||P(z_0)| - |P(z)||}{|z - z_0|}.$$

Since $|P(z_0)| \geq (R - \rho)^d = 1$ (for monic P) and $P(z_0) \neq 0$, the divided difference $Q(z_0, z)$ **never vanishes** for z_0 on the Cauchy circle, unlike $P'(z)$ which vanishes at critical points.

Remark 5.13 (Pandrosion vs. Newton: eliminating the Σ barrier). *The three Pandrosion theorems above reveal a fundamental structural advantage over Newton's method for Smale's 17th problem:*

	<i>Newton</i>	<i>Pandrosion</i>
Denominator	$P'(z)$	$Q(z_0, z) = (P(z) - P(z_0))/(z - z_0)$
At critical points	$P'(z) = 0$	$ Q \geq (R^d - (2\rho)^d)/(1 + 2\rho) \gg 0$
Sum identity	$\sum 1/w_k = 1$	$\sum 1/\omega_k = d - P'/Q \approx d$
Product bound	$\leq e^{-1+\Sigma/2}$ (conditional)	exact: $P(z_0) \prod Q_k/Q^d$
Σ condition	$\bar{\Sigma} < 2$ (unproved)	no analogue needed

5.6 The Pandrosion kinematic conservation law

Theorem 5.14 (Pandrosion Kinematic Identity). *Let $P \in \mathbb{C}[z]$ and z_0 an anchor with $P(z_0) \neq 0$. For any z not fixed by F_{z_0} :*

$$\frac{P(z)}{P(z_0)} = \frac{F_{z_0}(z) - z}{F_{z_0}(z_0) - z_0}. \quad (26)$$

Theorem 5.15 (Global Conservation Law). *Along any Pandrosion orbit $z_{n+1} = F_{z_0}(z_n)$:*

$$\prod_{n=0}^{N-1} \left(1 - \frac{P(z_n)}{P(z_0)} \right) = \frac{z_{\text{init}} - z_0}{z_N - z_0}. \quad (27)$$

If the orbit converges to a root ζ_k :

$$\prod_{n=0}^{\infty} \left| 1 - \frac{P(z_n)}{P(z_0)} \right| = \frac{|z_{\text{init}} - z_0|}{|\zeta_k - z_0|}. \quad (28)$$

5.7 The far-anchor obstruction and the scaling principle

Proposition 5.16 (Far-anchor obstruction). *Let $P(z) = z^d - 1$ with anchor z_0 on the Cauchy circle $|z_0| = R = 1 + \rho$. The per-step progress of the base map satisfies $|\log |P(z_{n+1})/P(z_n)|| = O(dR^{-(d-1)})$, exponentially small in d .*

Remark 5.17 (The scaling principle). *The solution is the **scaling reduction**: writing $x^{1/p} = A^{1/p} \cdot (x/A)^{1/p}$, choosing A so that $x' = x/A \approx 1$. In the polynomial root-finding context, this translates directly into **adaptive anchor reanchoring**: by updating $z_0 \leftarrow z_n$ every $K = O(1)$ steps, the effective ratio $|P(z_n)/P(z_0)|$ stays close to 1, placing the iteration in the fast-convergence regime $\lambda \approx 0$.*

5.8 Amortised block descent for the adaptive scheme

Definition 5.18 (Pandrosion- T_3 with period- K reanchoring). *The adaptive scheme operates in epochs. In each epoch with anchor z_0 :*

1. *Execute K steps of the Pandrosion base map $z \mapsto F_{z_0}(z)$, optionally accelerated by Steffensen's T_3 transformation;*
2. *Update the anchor: $z_0 \leftarrow z_K$.*

For $K = 1$, the scheme recovers Newton's method. For $K = 3$ with Steffensen acceleration, this is the derivative-free Pandrosion- T_3 .

Conjecture 5.19 (Amortised block descent — generic-polynomial form). *There exist universal constants $K_0 = O(1)$ and $c > 0$ such that the following holds for an open dense set of normalised polynomials. For any normalised polynomial $P \in \mathbb{C}[z]$ of degree d with simple roots in this set, and any reanchoring period $K \leq K_0$, the adaptive Pandrosion- T_3 scheme satisfies: for any block of K consecutive steps within a single epoch,*

$$|P(z_K)| \leq e^{-c} |P(z_0)|, \quad (29)$$

where the constant $c > 0$ is independent of d and P .

Remark 5.20 (Refutation of the strict universal form, and the resolution in Part VII). *The strict universal form of Conjecture 5.19 (asserting the bound for every normalised P) was refuted by computational search reported in Part VII [14]: there exist root constellations and internal anchors z_0 acting as exact stagnation traps, where the geometric sum of logarithms $\Sigma_{\log} > 0$*

and $|P(z_K)| \approx |P(z_0)|$. This refutation is consistent with McMullen’s topological impossibility theorem [11] for purely rational iterative schemes of degree $d \geq 4$. Part VII resolves the obstruction by introducing a derivative-free non-holomorphic fallback (Steffensen-Pandrosion-Armijo), whose descent is established as a quantitative theorem (Theorem 3.3 of [14]), and an analytic γ -damped variant (Theorem 4.2 of [14]), restoring the deterministic worst-case bound to $O(d^2 \log d)$. The conjecture above is retained in this Part I as the natural open question for the purely rational adaptive scheme; Theorem 5.27 below should be read as a conditional result on Conjecture 5.19, with the unconditional $O(d^2 \log^2 d)$ and $O(d^2 \log d)$ bounds appearing in Part VII [14].

5.9 The contraction ratio theorem

Theorem 5.21 (Pandrosion contraction ratio at fixed points). *Let $P \in \mathbb{C}[z]$ with simple roots $\zeta_1, \dots, \zeta_d$. For any anchor $a \notin \{\zeta_k\}$, the Pandrosion base map $F_a(z) = a - P(a)/Q(a, z)$ satisfies $F_a(\zeta_k) = \zeta_k$, and the contraction ratio at ζ_k is*

$$\lambda(a, \zeta_k) := |F'_a(\zeta_k)| = \left| 1 + \frac{P'(\zeta_k)(\zeta_k - a)}{P(a)} \right|. \quad (30)$$

In particular: (i) Cauchy circle regime ($|a| = R = 1 + \rho$, $|\zeta_k| \leq \rho$): $\lambda \rightarrow 1$ as $d \rightarrow \infty$. (ii) Scaling regime ($a \rightarrow \zeta_k$, $|a - \zeta_k| = \varepsilon \ll 1$): $\lambda = O(\varepsilon) \rightarrow 0$ (quadratic convergence).

Theorem 5.22 (Universal descent constant). *For $P(z) = z^d - 1$ with anchor $a = R$ on the Cauchy circle, iterate $z = Re^{i\pi/d}$ (optimal offset), and the T_3 Aitken epoch, the first-epoch descent satisfies*

$$\log \frac{|P(\hat{a})|}{|P(a)|} = -\frac{\pi}{2} - \frac{C}{d} + O(d^{-2}),$$

where $C \approx 0.530$. Equivalently, the epoch contraction ratio satisfies $\Lambda_{\text{epoch}} \rightarrow e^{-\pi/2} \approx 0.20788$.

5.10 Cluster domination with close anchors

Definition 5.23 (Dominant cluster). *Given $z \in \mathbb{C}$ and a scale $\sigma > 0$, the σ -dominant cluster is $I_\sigma(z) := \{k : |z - \zeta_k| \leq \sigma\}$.*

Theorem 5.24 (Local domination of cofactors — close-anchor regime). *Let z_0 and z satisfy $|z - z_0| \leq \delta$ for some $\delta > 0$, and suppose $|r| = |P(z)/P(z_0)| \leq 1/2$. Define $\Delta_k := (F_{z_0}(z) - z)/(z - \zeta_k)$. Then:*

1. For each k : $\Delta_k = r(z - z_0)/((1 - r)(z - \zeta_k))$, and $|\Delta_k| \leq 2|r|\delta/|z - \zeta_k|$.
2. For $|z - \zeta_k| \geq 4|r|\delta$: $|\log |1/\omega_k|| \leq 4|r|\delta/|z - \zeta_k|$.
3. Summing over roots with $|z - \zeta_k| \geq \sigma \geq 4|r|\delta$:

$$\sum_{k \notin I_\sigma(z)} |\log |1/\omega_k|| \leq 4|r|\delta \sum_{k \notin I_\sigma(z)} 1/|z - \zeta_k|.$$

5.11 Numerical verification on adversarial families

Proposition 5.25 (Block descent for the adaptive scheme). *Let $K = 3$ (with Steffensen T_3). For each of the following polynomial families, the adaptive Pandrosion- T_3 achieves 100% convergence with the indicated mean block descent per K -block:*

Family	d	Mean block descent	c (effective)
<i>Roots of unity</i>	100	−5.72	5.72
<i>Random circle</i>	100	−5.77	5.77
<i>Clustered line</i>	100	−5.34	5.34
<i>Roots of unity</i>	500	−5.39	5.39
<i>Random circle</i>	500	−5.38	5.38

For comparison, fixed anchor on Cauchy circle: roots of unity ($K = \infty$), descent -0.0003 .

5.12 Closing the counting argument

Corollary 5.26 (Basin entry from the adaptive scheme). *If Conjecture 5.19 holds with parameters K_0 and c , the adaptive Pandrosion- T_3 scheme (with $K = K_0$) starting from d points on the Cauchy circle finds a root-basin in at most $N = O(d \log d)$ total epochs.*

Theorem 5.27 (Conditional univariate complexity bound; relation to Smale 17). *If Conjecture 5.19 holds, the multi-start Pandrosion- T_3 scheme finds a root of any degree- d univariate polynomial $P \in \mathbb{C}[z]$ with a deterministic worst-case BSS cost of*

$$O(d^2 \log d) + O(d \log \log \varepsilon^{-1}). \quad (31)$$

The multivariate extension (Section 3.5) provides an analogous derivative-free algorithm for systems $F : \mathbb{C}^n \rightarrow \mathbb{C}^n$ with Bézout degree $D = d^n$, with empirical convergence on Smale-17-style Kostlan–Smale ensembles up to $(n, d) = (4, 3)$ (cf. Theorem 3.6). Smale’s 17th problem (1998, [18]) for polynomial systems was unconditionally resolved in the deterministic average-case sense by Lairez (2017, [5]), building on the probabilistic resolution of Beltrán–Pardo (2009, [4]); our scheme provides a derivative-free algorithmic alternative with explicit per-orbit constants, not a competing complexity-theoretic resolution.

Remark 5.28 (Roadmap toward proving Conjecture 5.19). *The strategy decomposes into three layers:*

1. **Scaling reduction** (Proposition 5.1): *reducing each epoch to the $x' \approx 1$ regime. Mechanism identified; formal reduction proved for $z^d - 1$.*
2. **Cluster domination** (Theorem 5.24): *far-root contributions are $O(\log d)$. Status: proved.*
3. **Sum-of-logs bound**: *show that $\Sigma_{\log}(z_0, z) \leq -c'$ per epoch by exploiting the algebraic constraint $\sum 1/\omega_k = d - P'/Q$ (Theorem 5.11) and the regularisation $|Q| \gg 0$ (Theorem 5.12). Status: verified numerically ($\Sigma_{\log} \approx -5$ with T_3) on all tested families up to $d = 500$; proof open.*

6 Supplementary results

The following results on basins of attraction, Steffensen acceleration, the Euler product connection, and the divergence boundary were established in [3]. We summarise them here for completeness.

6.1 Basins of attraction and fractal boundaries

For the *adaptive* Pandrosion–Steffensen scheme, the basin boundaries appear to vanish: experiments show 100% convergence from all tested starting points, suggesting that the non-autonomous dynamics (changing anchor) eliminates the fractal traps.

6.2 Steffensen acceleration in $\mathbb{C}$

Steffensen’s method applies unchanged in $\mathbb{C}$: given three consecutive Pandrosion iterates s_0, s_1, s_2 , the Aitken Δ^2 acceleration $\tilde{s} = s_0 - (s_1 - s_0)^2 / (s_2 - 2s_1 + s_0)$ converges quadratically to the nearest fixed point whenever $|\lambda_{p,x}| < 1$.

6.3 The Pandrosion–Euler connection

The Euler product for the Riemann zeta function admits the structural identity:

$$\zeta(s) = \prod_{p \text{ prime}} \frac{1}{1 - p^{-s}} = \prod_{p \text{ prime}} S_{\infty}(p^{-s}),$$

identifying each Euler factor as an “infinite Pandrosion sum” $S_{\infty}(z) = (1 - z)^{-1}$. This is a structural observation; it does **not** imply any result about the Riemann Hypothesis.

6.4 The divergence boundary

For $p = 2$, an explicit computation yields $|\lambda_{2,x}| = |x^{1/2} - 1| / |x^{1/2} + 1|$, which equals 1 on the imaginary axis, giving $\mathcal{D}_2 = \{\operatorname{Re}(x) \leq 0\}$. For $p \geq 3$, the divergence set shrinks: $\mathcal{D}_p \subsetneq \mathcal{D}_2$.

7 Conclusion

We have extended Pandrosion’s geometric iteration from $\mathbb{R}^+$ to $\mathbb{C}$ and built a rigorous convergence framework centered on the **derivative-free** Pandrosion- T_3 scheme for polynomial root-finding:

- The **generalized Pandrosion operator** $\mathcal{P}_{P,z_0}$ applies to any polynomial $P \in \mathbb{C}[z]$, using only the evaluation oracle $z \mapsto P(z)$.
- **Radial contraction** (Theorem 5.5): for $|z| > \rho$, the excess radius contracts as $e(N_P(z)) \leq (1 - 1/d)e(z)$.
- **Per-root contraction** (Theorem 5.7): on the Cauchy circle, every root is simultaneously approached. Verified with zero violations in 1,900,000 tests.
- **Newton product contraction** (Theorem 5.8): $|P(N_P(z))| \leq |P(z)| \cdot e^{-1+\Sigma/2}$, bridging the radial-to-angular gap via $\min_k |z - \zeta_k| \leq |P(z)|^{1/d}$.
- **Pandrosion product identity** (Theorem 5.10): the exact algebraic factorisation $P(F_{z_0}(z))/P(z) = P(z_0)/c_d \cdot \prod Q_k/Q^d$.
- **Pandrosion sum identity** (Theorem 5.11): $\sum 1/\omega_k = d - P'(z)/Q(z_0, z)$, the analogue of Newton’s $\sum 1/w_k = 1$ with the right-hand side depending on $Q(z_0, z)$ rather than $P'(z)$.
- **Regularisation** (Theorem 5.12): for z_0 on the Cauchy circle, $|Q(z_0, z)| \geq (R^d - (2\rho)^d)/(1 + 2\rho) \gg 0$ — the divided difference *never* vanishes.
- **Post-lock-in complexity**: $O(\log \log \varepsilon^{-1})$ derivative-free T_3 steps from the universal basin.
- **Scaling principle** (Proposition 5.1, Remark 5.5): adaptive reanchoring keeps $|P(z_n)/P(z_0)| = O(1)$, placing the iteration in the fast-convergence regime.
- **Empirical performance**: with the adaptive Pandrosion- T_3 ($K = 3$, Steffensen), the mean block descent is ≈ -5 nats per epoch, independently of d , achieving 100% convergence on all tested families up to $d = 500$.

What remains open. Theorems 5.10–5.12 provide an exact product factorisation and a bounded divided difference, eliminating the conditional $\bar{\Sigma} < 2$ barrier that blocks Newton-based proofs. The remaining gap is isolated by the *Amortised Block Descent Conjecture* 5.19: each K -epoch must reduce $|P|$ by a factor e^{-c} with $c > 0$ universal. The roadmap (Remark 5.20) decomposes this into three layers: scaling reduction (proved for $z^d - 1$), cluster domination (proved), and sum-of-logs bound (verified numerically; proof open). If confirmed, this yields basin entry in $O(d \log d)$ epochs and gives a derivative-free $O(d^2 \log d)$ -operation BSS algorithm for univariate root-finding (Theorem 5.27). The multivariate extension to systems (Section 3.5) provides the genuine Smale-17 algorithmic framework, complementing the deterministic average-case resolution of Lairez [5] with a derivative-free oracle.

Perspectives.

- *Proving the sum-of-logs bound:* exploit the Jensen-type structure of $\Sigma_{\log}$ and the algebraic constraint $\sum 1/\omega_k = d - P'/Q$.
- *Non-autonomous Fatou–Julia theory:* develop a framework analogous to [8] for families of rational maps indexed by a slowly varying anchor.
- *Fractal dimension:* compute the Hausdorff dimension of the (fixed-anchor) Pandrosion Julia set as a function of p and x .
- *Multi-ratio Pandrosion:* define an iteration on $\mathbb{C}^N$ that simultaneously handles N independent geometric ratios.
- *Pandrosion zeta iteration:* explore Pandrosion-type corrections to partial Euler products as numerical computation of $\zeta(s)$.
- *p -adic extension:* the geometric sum S_p makes sense over p -adic fields.

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